

1 (a)

$$S_y = u_y t + \frac{1}{2} a t^2$$

$$-1.0 = u_y(0.80) + \frac{1}{2}(-9.81)(0.80)^2 \quad [\text{M1- correct } S_y \text{ and } a \text{ values}]$$

$$u_y = 2.674$$

$$= 2.67 \text{ m s}^{-1} \quad [\text{A0}]$$

(b)

$$\frac{u_y}{u_x} = \tan(10^\circ)$$

$$u_x = \frac{2.67}{\tan(10^\circ)} \quad [\text{M1}]$$

$$= 15.1 \text{ m s}^{-1} \quad [\text{A1}]$$

(c)

$$S_x = u_x t$$

$$S_x = (15.1)(0.80) = 12.1 \text{ m} \quad [\text{C1- for calculating best value}]$$

$$\frac{\Delta S_x}{S_x} = \frac{\Delta u_x}{u_x} + \frac{\Delta t}{t}$$

$$= \frac{5.0}{100} + \frac{0.08}{0.80} = 0.15 \quad [\text{C1: correct substitution}]$$

$$\Delta S_x = 0.15(S_x) = 1.8 \text{ m}$$

$$S_x \pm \Delta S_x = 12 \pm 2 \text{ m} \quad [\text{A1}]$$

(d) (i) Air resistance acts in the same direction/ downward as the weight of ski jumper. [B1]

Hence, the vertical acceleration of the ski jumper is greater than that of free fall or net force is downwards. [B1]

Air resistance decreases with decreasing speed. Hence magnitude of vertical acceleration decreases. [B1]

At highest point, vertical acceleration is g or free fall (as there is no resistance). [B1]

Maximum 3 marks for any of points stated above.

(ii) This position increases her surface area to generate upwards lift force.

The more the lift, the more time she stays on the flight and thus, the further she will travel. [B1]

Or:

This position decreases her exposed surface area horizontally (or more streamlined/ reduce the air resistance). Hence, she can travel more horizontally.



2

Or:

This position will lower her CG such that there is a counter moment to the moment created by air resistance.